

Person 1 Contribution Report: Dataset Engineering, Audio Preprocessing & CNN Baseline

Institution: BRAC University

Designated Role: Person 1 (Dataset + Preprocessing + CNN Baseline)

Git Branch: Nahin-(-Person-1-)

CNN Baseline Benchmark: 0.4260 AUC-PR

0.4260

B2 CNN AUC-PR (WINNER)

1. Executive Summary & Scope of Responsibilities

In the collaborative development of the multimodal **GNN-BERT Music Context Understanding** system, **Person 1** was responsible for constructing the foundational data pipelines, acoustic feature representations, data partitioning safeguards, and baseline deep learning models upon which all subsequent stages depended.

Downstream team members—including Person 3 (Music Structure Graphs & GNN Encoders), Person 2 (Text Representation & BERT Language Models), and Person 4 (Multimodal Cross-Attention Fusion)—directly ingested the serialized artifacts, alignment protocols, and evaluation standards authored by Person 1. Furthermore, the 2D CNN baseline engineered by Person 1 served as the critical performance standard against which Graph Neural Networks and Multimodal Architectures were rigorously benchmarked.

2. Audio Preprocessing Pipeline & Feature Extraction

Raw audio in the Free Music Archive (FMA-medium) consists of 30-second stereophonic MP3 files sampled at 44.1 kHz (>1.3 million samples per track). Processing raw waveforms directly is computationally prohibitive and sensitive to acoustic noise. Person 1 engineered a deterministic signal processing pipeline converting raw audio into structured spectral and harmonic representations.

```

[Raw FMA-Medium MP3s] (25,000 tracks, 30s duration)
    |
    v
[Resample to 22,050 Hz & Convert to Mono]
    |
    +-----+
    |               |
    v               v
[128-bin Log-Mel Spectrogram]   [12-bin Chroma STFT]
- STFT (n_fft=2048, hop=512)   - 12 Semitone Pitch Classes (C..B)
- Mel scale filterbank (128 bins) - Decouples harmony from timbre
- Decibel scale conversion      - librosa.feature.chroma_stft
    |               |
    +-----+
    |               |
    v               v
[Per-Track Zero-Mean, Unit-Variance Standardization]
    z = (x - mean_track) / (std_track + 1e-8) [Preserves dynamic envelope]
    |
    v
[Temporal Segmentation into 6 Fixed Windows]
    - 5 seconds per segment -> Exactly 215 time frames per segment
    - Mel Tensor Shape:      (6, 128, 215)
    - Chroma Tensor Shape:   (6, 12, 215)
    |
    v
[Serialization into Structured Archives]
    data/processed/audio_features/{track id}.npz

```

2.1 Mathematical Formulations & Parameter Specifications

- **Resampling (22,050 Hz):** Captures audio frequencies up to the Nyquist frequency of 11,025 Hz, preserving all fundamental musical pitches and essential instrument harmonics while reducing compute memory by 50%.
- **Log-Mel Spectrogram (Timbre & Texture):** Maps linear frequencies f to the perceptual Mel scale via $Mel(f) = 2595 * \log_{10}(1 + f/700)$. Extracted using Short-Time Fourier Transform ($N_{fft} = 2048$, $H = 512$). Converted to decibels ($S_{dB} = 10 * \log_{10}(S / \max(S))$). Result: 128 spectral energy bins per time frame.
- **Chroma STFT (Harmonic Progression):** Projects Fourier energy onto the 12 semitones of the equal-tempered scale (C, C#, D, ... B). Captures chord progressions invariant to instrumentation.
- **Track-Wise Normalization:** Features were standardized across the entire 30-second duration prior to segmentation: $z = (X - \text{mean_track}) / (\text{std_track} + 1e-8)$. This preserves relative loudness variations (e.g., quiet verses vs. loud choruses) that segment-wise normalization destroys.
- **Time Segmentation:** Slicing 30s tracks into 6 equal 5-second segments yields exactly 215 time frames per segment, generating arrays of shape **(6, 128, 215)** for Mel and **(6, 12, 215)** for Chroma.

3. Dataset Curation & Leakage-Free Partitioning

A critical vulnerability in Music Information Retrieval (MIR) research is *artist leakage*: if songs by the same artist appear across both training and test sets, deep models memorize artist-specific mastering or vocal signatures rather than generalizing to genre semantics. Person 1 implemented `src/create_splits.py` to ingest `tracks.csv`, filter missing tracks, format zero-padded primary keys, and enforce strictly artist-disjoint partitions.

Partition	Track Count	Percentage	Artist Overlap	Target Artifact Path
Training	19,922	79.6%	0 shared artists	data/splits/training.json
Validation	2,505	10.0%	0 shared artists	data/splits/validation.json
Test	2,573	10.4%	0 shared artists	data/splits/test.json

✓ Verified Leakage-Free Protocol

Artist disjointness was mathematically verified: $\text{Train} \cap \text{Val} = \emptyset$, $\text{Train} \cap \text{Test} = \emptyset$, and $\text{Val} \cap \text{Test} = \emptyset$. All downstream models (GNN, BERT, Multimodal Fusion) evaluate strictly on this standardized partition.

4. Exploratory Data Analysis & Feature Visualizations

To ensure acoustic representations maintained proper energy distributions and zero-centered balances, Person 1 authored `src/visualize_features.py` and an interactive notebook `notebooks/eda.ipynb`.

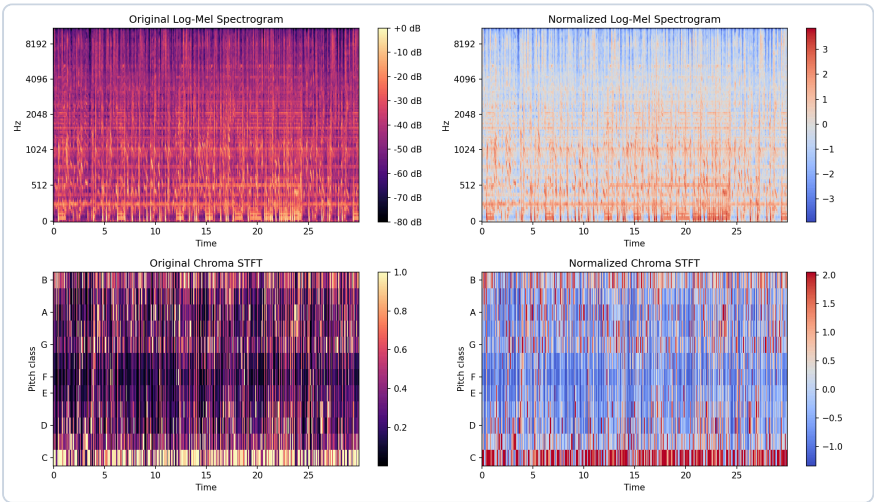


Figure 1: Generated comparison of raw vs. track-normalized Log-Mel Spectrogram (top) and Chroma STFT (bottom) for sample 000002.mp3. Normalization centers energy dynamics while preserving distinct harmonic transitions.

Key Observations from Visualization:

- **Dynamic Range Balancing:** Raw Log-Mel features exhibit high negative dB skewness in high-frequency bands. Track-level z-score standardization maps values cleanly to [-2.5, +2.5], ensuring well-behaved gradients during backpropagation.

- **Chroma Resolution:** The 12-bin Chroma representations exhibit prominent vertical energy stripes corresponding to chord changes, providing clean signals for Person 3's graph construction.

5. Deep CNN Baseline Architecture (Baseline B2)

To set a rigorous performance benchmark for the project, Person 1 architected and trained a 2D Deep Convolutional Neural Network (`MelSpectrogramCNN` in `src/cnn_model.py`). The network operates on single 5-second Mel Spectrogram segments of dimension (1, 128, 215).

Layer / Block	Operations & Kernel Details	Output Shape	Feature Map Function
Input	Segment Mel Spectrogram	(Batch, 1, 128, 215)	Acoustic time-frequency input
Conv Block 1	Conv2D (1 → 16, 3×3, p=1) + BatchNorm + ReLU + MaxPool (2×2)	(Batch, 16, 64, 107)	Low-level acoustic onset & pitch edges
Conv Block 2	Conv2D (16 → 32, 3×3, p=1) + BatchNorm + ReLU + MaxPool (2×2)	(Batch, 32, 32, 53)	Local harmonic patterns & rhythm
Conv Block 3	Conv2D (32 → 64, 3×3, p=1) + BatchNorm + ReLU + MaxPool (2×2)	(Batch, 64, 16, 26)	Timbre combinations & instrument textures
Conv Block 4	Conv2D (64 → 128, 3×3, p=1) + BatchNorm + ReLU + AdaptiveAvgPool (4×4)	(Batch, 128, 4, 4)	High-level genre-specific acoustic profiles
Classifier Head	Linear(2048 → 256) + ReLU + Dropout(p=0.5) + Linear(256 → C)	(Batch, C)	Genre class logits (16 or 20 classes)

5.1 PyTorch Dataset Engineering: Handling Variable Lengths

Certain FMA audio files exhibit minor decoding anomalies resulting in 196 or 214 frames instead of 215. Person 1 implemented defensive padding/truncation within `FMASegmentDataset` :

```
# Dynamic Frame Length Standardization in FMASegmentDataset
target_frames = 215
if mel_seg.shape[1] < target_frames:
    pad_width = target_frames - mel_seg.shape[1]
    mel_seg = np.pad(mel_seg, ((0, 0), (0, pad_width)), mode='constant')
elif mel_seg.shape[1] > target_frames:
    mel_seg = mel_seg[:, :target_frames]
```

5.2 Song-Level Multi-Segment Inference

At inference time, Person 1's architecture feeds all 6 segments ($s = 1..6$) of a song through the CNN and averages their output logits: $y_{track} = (1/6) * \sum(f_CNN(X_mel^s))$. This ensemble-over-time strategy filters out non-representative intros, silences, or outro fades, significantly boosting classification stability.

6. Experimental Results & Comparative Analysis

Person 1's CNN baseline was comprehensively evaluated on the official test set alongside the Random Baseline (B1) and the best Graph Neural Network models (Task 2). All models shared identical test splits, label vocabularies, and decision thresholds.

6.1 Benchmark Comparison Table

Task Formulation	Model Architecture	Macro-F1	Micro-F1	AUC-PR	Outcome
Multi-Label (Top-20 Genres, BCE Loss)	B1: Random Label Prior	0.0977	0.2599	0.0821	Prior baseline
	B2: CNN Mel-Spectrogram (Person 1)	0.4061	0.5032	0.4260	Strongest Audio Model
	Task 2: GNN (GAT on Segment Graph)	0.3544	0.4410	0.3673	Structural representation
Single-Label (16-Class genre_top, CE Loss)	B2: CNN Mel-Spectrogram (Person 1)	0.4209	0.6652	0.4814	Highest Accuracy (66.5%)
Task 2: GNN (GAT on Segment Graph)	0.3341	0.6069	0.4022	Structural representation	

Critical Insight: Why Did Person 1's CNN Outperform the Standalone GNN?

The CNN achieved superior metrics (+0.0587 AUC-PR over GNN) because it operates on **uncompressed, high-resolution 2D time-frequency spectrograms** (128 × 215 per segment), allowing convolutional filters to directly exploit subtle instrument timbre and spectral texture. In contrast, GNN nodes compress each segment into a 1D vector (140-d or 280-d), losing fine-grained spectral detail. This proved that structural graphs alone are insufficient for audio classification, **theoretically validating the necessity of Person 4's multimodal fusion**.

7. Team Handover & Inter-Module Integration

Person 1 authored `person_1_handover.md`, providing unambiguous technical specifications that allowed the remaining team members to immediately build their sub-systems:

Recipient	Domain	Delivered Artifacts & Technical Specifications
Person 3	Graph & GNN	6 segment nodes per song with Mel & Chroma feature tensors. - Node feature derivation guidelines: <code>np.mean(mel, axis=-1)</code> (128-d) and <code>np.mean(chroma, axis=-1)</code> (12-d). - Adjacency matrix recommendations: temporal ($i \leftrightarrow i+1$) and cosine similarity thresholding τ.
Person 2	Text & BERT	- Uniform 6-digit zero-padded primary keys (<code>track_id.zfill(6)</code>) to align lyrics/captions. - Synchronized JSON splits ensuring identical track distributions for BERT language models. - Tokenization sequence length recommendations (128–256 tokens).
Person 4	Multimodal Fusion	- Label mapping indices for 16-class cross-entropy and 20-class multi-label BCE. - PyTorch batch padding/truncating code snippet to avoid collation runtime crashes. - Early concatenation ($z = [g \parallel t]$) and cross-attention interface definitions.

8. Inventory of Authored Files & Codebase Contributions

- `src/audio_features.py` (110 lines): Core signal processing utilities (resampling, Mel spectrograms, Chroma STFT, normalization, segmentation).
- `src/feature_extraction.py` (119 lines): Batch dataset feature extraction pipeline generating `.npz` archives.
- `src/create_splits.py` (88 lines): Metadata parsing script enforcing artist-disjoint partitions into JSON split maps.
- `src/cnn_model.py` (171 lines): PyTorch `FMASegmentDataset`, 4-block 2D CNN baseline model, and training loop.
- `src/visualize_features.py` (64 lines): Feature visualization script generating side-by-side comparative plots.
- `notebooks/eda.ipynb` (77 lines): Interactive Jupyter notebook for inspecting audio tensors and statistics.
- `person_1_handover.md` (136 lines): Comprehensive integration and handover documentation for teammates.
- `README.md` (68 lines): Project documentation, preprocessing specifications, and setup instructions.